

Conformal Prediction as a Transform within a Grammar for Probabilistic Forecasting

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Abstract

Conformal prediction is typically presented as a bundle: a base model, a scalar residual, and a rank-calibration step carrying an average coverage guarantee. The dangers of chasing coverage alone are just as well documented. Building an autonomous economic prediction system forced a different arrangement on us: a grammar of distributional transforms in which one univariate transform embodies the conformal step but need not terminate the processing. Its peers are the classical tools of time-series analysis, all state-conditional bijections. This resolves the inconsistency of treating residuals unconditionally beneath a conditional base model, and keeps the coverage certificate where wanted, though it does not propagate down the chain. On the economic series studied, the conformal step earns its place under both log-likelihood and CRPS, though a fixed heavy-tailed link captures most of the same benefit without the certificate.

1 Introduction

In the canonical residual-pooling pattern, a forecaster fits a conditional model and then quotes uncertainty by ranking its past errors with every state pooled. The ranking is honest on average and silent about tonight. That tension, conditional effort upstream and unconditional treatment at the end, is the subject of this paper.

Split conformal prediction can be written as a chain. Fit a model, take residuals, apply the fence-post empirical map

$$E_n(r) = \frac{1 + \#\{i \leq n : R_i \leq r\}}{n + 1}, \quad (1)$$

and map the resulting rank back through the model to a prediction set (Vovk et al., 2005; Lei et al., 2018). The value of (1) at a new point is a conformal p-value, and the map itself is the unrandomized conformal predictive distribution of Vovk et al. (2019).

We call it the fence-post transform, after the two +1 corrections: n points cut the line into $n + 1$ cells, and the guarantee rides on the count of cells, not of points. The knot probabilities $i/(n + 1)$ are Weibull's plotting positions, the means of uniform order statistics, so the object is classical.

The literature mostly treats this chain as a construction to be studied whole, and its many variants, normalized, Mondrian, quantile, distributional, and temporal (Boström et al., 2021; Romano et al., 2019; Chernozhukov et al., 2021; Gibbs and Candès, 2021; Auer et al., 2023), as separate constructions. The nearest exceptions stop one step short. Conformal predictive

systems made the map emit a distribution function. Conformal calibrators (Vovk et al., 2020) apply that map to another system’s output. Both compose once, and the map is still the last word.

We propose a different organization, in which the map’s position is a variable. The fence-post transform is one operator. The chain is one production rule in a grammar of composable transforms. The interesting question is then not which conformal method to choose but which sequence of transforms minimizes the forecaster’s scoring rule.

Viewed this way, conformal prediction is not, in general, an alternative that puts statistical modeling to one side, but an integrated operator in a chain, whose finite-sample properties may well be valuable at the point where they are computed. Those properties distinguish it from the operators it commingles with, but they do not privilege stopping there, nor need they remove the modeling that follows. Used carefully in this manner, it does not materially degrade predictive quality under a proper scoring rule, at least in our experiments.

The standard conformal pattern also carries a tension the present work avoids. A skeptic of the standard conformal prediction patterns might reasonably ask why heavy anchoring on the empirical distribution (in a sense, not modeling at all) is appropriate in the context of model residuals, whereas to apply the same emphasis in general would be absurd.

The stock answer is about targets. For marginal coverage, pooling is justified by exchangeability of the residuals alone, a weaker and testable hypothesis, and distribution-free conditional coverage is provably out of reach (Vovk, 2012; Lei and Wasserman, 2014; Foygel Barber et al., 2021), so the pooled law is what one settles for.

That answer is not satisfactory for a forecaster scored by a proper scoring rule. There the pooled residual law is quoted as a predictive density, its adequacy is exactly conditional invariance, and the certificate does not establish that. Certifying ranks under exchangeability makes stopping at the pooled law a modeling decision, not a consequence of the guarantee.

In univariate probabilistic forecasting, Platt scaling, isotonic and quantile recalibration, temperature scaling, conformal predictive systems, and distributional conformal prediction can all be represented as monotone recalibration operators on a scalar predictive, score, or probability coordinate, differing in what they fit and what they guarantee (Section 5). The fence-post transform is the member with a certificate, and the certificate is a positional type annotation on one operator inside a composable predictive program: a reason to admit the operator to the grammar, and nothing more.

This reorganization has content because the fence-post operator composes. Once interpolated, E_n is an invertible change of variables with a density, and its probit composition $z = \Phi^{-1}(E_n(r))$ maps exchangeable residuals to approximately standard Gaussian coordinates. Nothing obliges a forecaster to stop there. One can fit further structure on the Gaussianized stream, transform again, and continue, exactly as one does with any other member of the transform library. Conformal calibration stops at the transform. The grammar keeps going.

Two testable claims follow. The first is that the operator is not a natural stopping point: a serial model fitted on the Gaussianized stream can beat quoting the transformed empirical law where it stands, when exploitable residual structure remains. The second is that nesting is cheap: a pool over chains that includes the conformal pattern should not pay materially for admitting it, because the pool reallocates weight away from the pattern on series where it is a bad choice. We confirm both on a corpus of economic series and report a third finding we did not anticipate: the operator’s measured value is dominated by whether any peer in the chain already bounds the emitted coordinate. An ablation with a self-normalized scale rung, which

bounds on the side, absorbs nearly all of the operator’s contribution, so the grammar’s type contracts are the empirical story, not only the formalism.

A companion paper prices what a fixed representation choice costs (Cotton, 2026b). Under logarithmic loss the irreducible cost of pooling residuals in given coordinates is a mutual information, the conformal information gap, and the companion measures it on the same corpus. The present paper is the constructive complement. If a fixed conformal pattern can be a bad choice with a quantifiable price, the remedy is not a better fixed choice. It is a grammar in which the choice is made by the scoring rule, online, per series.

2 The fence-post transform as an operator

We work with online univariate transforms in the sense of the `skaters` library, described at length in a methods paper (Cotton, 2026d) and briefly in a software paper (Cotton, 2026e). A transform is a pair of functions. The forward function consumes one observation and its own past state and emits a transformed observation. The inverse function maps predictive distributions in the transformed space back to the original space. Conditional on its own past state, the forward map is an invertible change of variables in the current observation, so predictive densities push back exactly, Jacobians included.

The raw fence-post map (1) is a step function. It has no density and no inverse, so it is not a transform in this sense. The minimal repair is interpolation, minimal but not unprecedented (Harar et al., 2022). Let $y_{(1)} \leq \dots \leq y_{(n)}$ be the past observations, place knots at the order statistics with fence-post probabilities $p_{(i)} = i/(n+1)$ for $i = 1, \dots, n$, set $z_{(i)} = \Phi^{-1}(p_{(i)})$, and interpolate the pairs $(y_{(i)}, z_{(i)})$ piecewise linearly. We call the resulting operator *gaussianize*. The knot values are the left limits of the right-continuous map (1) at the calibration points, and this convention is what makes the certificate exact at the knots (Proposition 5). That is the full map. The implementation retains at most 39 of the knots, chosen at equal mass from the full sorted history so that each retained knot is an exact order statistic until the slope guard below binds, and drops any knot whose value ties its neighbor, so ties widen the retained cells. We write *gaussianize-full* for the map through every knot and *gaussianize* for the implemented retained-knot member. Proposition 7 prices the difference: the certificate is a function of the retained resolution.

The interpolation changes the object, and Section 3 states exactly what survives the change: the discrete map’s output is a function of ranks alone, while the interpolated map’s output depends on spacings as well, so the two are different operators with different certificates.

Beyond the edge knots the map extends linearly in (y, z) coordinates, and linearity in the probit coordinate is a Gaussian tail in the observable, with scale set by the spacing of the extreme quantiles. Tail mass is unobservable, so any usable version of the operator carries a tail assumption, and this is the mildest one consistent with the coordinates. A kernel-smoothed variant replaces the empirical distribution function with a Gaussian-kernel estimate at Silverman bandwidth, represented by a monotone C^1 cubic through the same knots, and Section 4.4 checks that smoothness changes little.

Two guards complete the definition, and both were forced by data rather than taste. First, every knot segment’s inverse slope dy/dz is capped at twenty times the interquartile inverse slope, so the forward slope is bounded below. An upstream variance collapse can plant a many-sigma outlier as an edge knot, and an uncapped edge segment then hands the inverse map an explosive extrapolation slope. The cap is an explicit assumption: tail scale at most a fixed

multiple of the central scale.

Second, the value handed downstream is clamped to $|z| \leq 8$. The clamp applies only to the coordinate supplied to downstream state updates and is not part of the invertible map used to construct or pull back predictive densities, so the change-of-variables identity is intact. When it binds, the downstream learner is updated with a bounded surrogate rather than the literal transformed observation, and that alteration can persist through the learner’s state. In the type system the two objects are distinct: gaussianize is invertible and density-carrying, and the bounded update channel is a winsorized learner input, not a transform.

Within the knot range the clamp cannot bind, since no fence-post rank exceeds probability $n/(n+1)$ and $\Phi^{-1}(n/(n+1)) < 8$ for any feasible n . It can bind on the extended tail and on degenerate warmup emissions, and the latter is why it exists, because downstream learners have expanding memory. In one series in our corpus a degenerate warmup emitted a single transformed value near 8000, and the downstream Welford variance and recursive least-squares state never forgot it, inflating that series’ loss by four orders of magnitude. Composability is not free. An operator that is harmless as a terminal step can poison downstream learners through one bad emission, and the operator’s definition has to prevent this, not the caller.

The guards are better read as declared budgets than as patches. Invertibility alone is too weak a property for a chain. The pullback quantile $y_\alpha = m^{-1}(q(\alpha))$ inherits the steepness of the inverse map m^{-1} , and beyond the edge knots the inverse is an assumption rather than an estimate, because tail mass is unobservable at any n . The slope cap makes the assumption finite. It declares that m^{-1} is Lipschitz, with constant L equal to twenty times the central inverse slope, and the declaration buys a guarantee. The declaration also has a price: a binding cap moves knots off their order statistics, and Appendix A prices the displacement.

Lemma 1 (Pullback quantiles are budgeted). *Let m be a strictly increasing forward map whose inverse is L -Lipschitz on all of $\mathbb{R}$, extrapolation included, and let q be the quantile function of the predictive in transformed coordinates. Then the pulled-back predictive has quantiles $y_\alpha = m^{-1}(q(\alpha))$ and*

$$|y_\alpha - y_{1/2}| \leq L |q(\alpha) - q(\tfrac{1}{2})| \quad \text{for every } \alpha \in (0, 1).$$

For a chain of such maps the same bound holds with L the product of the rungs’ constants.

Proof. Monotone maps transport quantiles, so $y_\alpha = m^{-1}(q(\alpha))$. The Lipschitz bound applied to the pair $q(\alpha), q(\frac{1}{2})$ gives the display. A composition of Lipschitz maps is Lipschitz with the product constant. $\square$

With a Gaussian leaf of scale σ the right side is $L\sigma |\Phi^{-1}(\alpha)|$. The budget is conditional on the operator’s state: L is twenty times the current central inverse slope and σ is fitted, so the lemma bounds quoted quantiles relative to the current central scale, not by a universal constant. Within that reading, no realization amplifies a quoted quantile past its budget. The lemma budgets the ideal pulled-back predictive. The implemented nine-particle mixture approximates that predictive, and the approximation error is not bounded here. The global L is also purchased with the knot displacement of Remark 10: the Lipschitz budget and retained-level exactness are competing constraints, and no operator holds both exactly on arbitrary data. The naive empirical map declares $L = \infty$ and buys nothing. The clamp is the same discipline in the forward direction. The retained knot coordinates lie within $(-8, 8)$ for every practical history length, so the clamp controls only extrapolated and warmup emissions, and a downstream learner with expanding memory is safe against exactly the emissions it excludes. Any finite multiplier yields

the lemma, so the value twenty is the one remaining choice of taste, a tail-conservatism dial. The budget statement is for the piecewise-linear operator: the Fritsch–Butland tangents of the smooth variant can steepen the inverse of a capped secant by a factor approaching two, so that variant’s declared budget is $2L$. Fitting the tail where the sample reaches, generalized-Pareto by censored maximum likelihood as the leaf already does, with the budget kept as a prior bound, is the refinement we leave open (the tail instability of the normal quantile transform is documented by Bogner et al., 2012).

Before enough observations accumulate, the operator standardizes by expanding moments, which is the $n \rightarrow 0$ limit of the empirical map. After the warmup the knots are rebuilt each step, at most thirty-nine of them at equal mass from the full sorted history, so the retained knots sit at exact order statistics, and carry no rank-approximation error while the slope guard is inactive. The sorted buffer itself grows with the series. A bounded compacted sketch is the obvious refinement, and its price is the rank-error term of Proposition 7.

3 What composition preserves

A *chain* is a finite word $C = (T_1, \dots, T_m, L)$, stateful transforms followed by a terminal leaf. A transform is a tuple: a forward emission, an inverse pullback, a state update, a clock, input and output types, and declared properties. The forward map consumes an observation and the operator’s state and emits a coordinate. The pullback carries predictive laws backward. The clock says when the state may absorb data. The properties are judgments, monotone, invertible, density-carrying, inverse-Lipschitz with a declared budget, exchangeability-preserving, previsible, or δ -calibrated at a position.

Composition obeys checkable rules. Invertible composed with invertible is invertible, and Lipschitz budgets multiply (Lemma 1). A fixed strictly increasing link transports the grid certificate, with its levels transported along the link, because ranks are invariant under it. Previsible composed with previsible is previsible, since the composite map at time t is built from maps measurable before t . A calibration certificate is not, in general, inherited through a fitted state-dependent operator, which is the second fact below. A pool over chains is itself an operator on chains, and Proposition 4 is its admission rule.

Composition preserves some properties and destroys others, and the pattern is positional. We record the four facts that organize our experiments.

Proposition 2 (Fence-post uniformity). *Let $R_1, \dots, R_{n+1}$ be exchangeable and almost surely distinct. Then $E_n(R_{n+1})$ is uniform on the grid $\{k/(n+1) : k = 1, \dots, n+1\}$. In particular, $\Pr\{E_n(R_{n+1}) \leq k/(n+1)\} = k/(n+1)$ for $k = 1, \dots, n+1$.*

Proof. By exchangeability every ordering of $R_1, \dots, R_{n+1}$ is equally likely, and by almost-sure distinctness the rank of R_{n+1} among the $n+1$ values is uniform on $\{1, \dots, n+1\}$. The map (1) evaluated at R_{n+1} equals the rank divided by $n+1$. $\square$

This is the standard conformal fact (Vovk et al., 2005) stated as a property of the operator’s output. It certifies the discrete map (1), and the discrete map is what the experiments do not run. The interpolated operator of Section 2 uses spacings as well as ranks, and its output between knots is not distribution-free. What survives interpolation is the grid, and the accounting is mechanical. Call an operator δ -conformal if, on exchangeable almost surely distinct input, its probability-scale output is within δ of calibrated at every level. Any interpolation through every

knot is $1/(n+1)$ -conformal, exact at the knots. The implemented 39-knot map is conformal to roughly one part in forty, exact at its retained levels while its guard sleeps. Compression, a binding guard, and ties each pay a price. Appendix A states and proves the prices, a leave-one-out version for data-dependent guards among them, and a counterexample where the moving cap dominates. The certificate is priced in the operator’s computational budget, and none of the prices moved the campaign’s scores.

What separates them is the exact statement. The discrete map is deterministic, exactly calibrated at every grid level, noninvertible, and $1/(n+1)$ -calibrated between grid levels. Randomized smoothing, which spreads each output uniformly over its rank cell, is exactly calibrated at every level and is a stochastic kernel rather than a deterministic transform (Vovk et al., 2005). The interpolated map is deterministic and invertible, stays exact on the grid, and incurs at most one full-grid cell of calibration error elsewhere. The grammar uses the deterministic, invertible version and quantifies its one-cell calibration relaxation.

The exact conformal object is a rank certificate, and a rank certificate is not a predictive density. Turning it into one requires interpolation, smoothing, tail assumptions, and model selection, the same choices met everywhere else in probabilistic forecasting, and exchangeability supplies none of them.

The log score will still notice: it integrates the predictive density between grid points and in the tails, exactly where the certificate is δ -relaxed and assumption-laden, which is where the fixed-link comparisons of Section 4.2 bite.

The second fact is that this guarantee is positional. Proposition 2 concerns the operator’s immediate output. If a downstream model is fitted on $z_1, \dots, z_t$ and its parameters enter later forecasts, the composed system’s forecast at time $t+1$ depends on the ordering of the past stream. The exchangeability that delivered uniformity no longer applies to the composed output. The calibration certificate attaches to a position in the chain, not to the chain. This is not a defect. It is the familiar price of fitting, and the reason the companion paper’s ledger separates representation, model-form, and estimation cost (Cotton, 2026b). But it means the grammar has a typing discipline: an operator’s guarantee names the position at which it holds, and claims about the composed system need their own arguments.

The third fact concerns schedules, and it explains what the conformal literature’s data splitting is for and how an online chain does without it. Call a chain *previsible* if the map applied to each observation is measurable with respect to information strictly prior to that observation. Dual use, the defect the splitting exists to avoid, is exactly a previsibility violation: an observation flowing through a map whose state already contains it.

Split conformal enforces previsibility with a coarse clock. Interior operators learn on a fitting block and freeze, the terminal fence-post learns on a disjoint calibration block and freezes, and test points meet only frozen maps.

The chains of this paper enforce the same rule with the finest clock. Every operator emits before it absorbs, so the map applied at time t is built from $y_1, \dots, y_{t-1}$ alone, and the past is a calibration set that never contains the present. Emit-before-absorb enforces non-anticipation, and nothing else. It does not preserve calibration-test exchangeability under adaptive upstream transforms, and the narrowing after Proposition 3 makes the difference from splitting precise.

On the exchangeable hypothesis the fine clock matches the batch guarantee and adds independence across time.

Proposition 3 (Sequential fence-post outputs are independent). *Let $Y_1, \dots, Y_T$ be exchangeable and almost surely distinct, and let $U_t = E_{t-1}(Y_t)$ be the fence-post transform of Y_t through the*

map built from $Y_1, \dots, Y_{t-1}$, as in (1) with $n = t - 1$. Then $U_1, \dots, U_T$ are independent, and U_t is uniform on $\{1/t, \dots, t/t\}$.

Proof. Let $K_t \in \{1, \dots, t\}$ be the rank of Y_t among $Y_1, \dots, Y_t$. From (1) with $n = t - 1$, $U_t = (1 + \#\{i < t : Y_i \leq Y_t\})/t = K_t/t$. The full ranking of $Y_1, \dots, Y_T$ is a permutation π of $\{1, \dots, T\}$, and the map $\pi \mapsto (K_1, \dots, K_T)$ is a bijection onto $\{1\} \times \{1, 2\} \times \dots \times \{1, \dots, T\}$: given the sequential ranks, the permutation is reconstructed by inserting each element at its stated position, and both sets have cardinality $T!$. Exchangeability with almost-sure distinctness makes π uniform over the $T!$ orderings, so $(K_1, \dots, K_T)$ is uniform over the product set. Uniformity over a product set is independence of the coordinates, each uniform on its factor. $\square$

The independence of sequential ranks is classical (Rényi, 1962) and powers online exchangeability testing (Vovk et al., 2003). The proposition concerns the discrete sequential ranks alone. The interpolated coordinates carry spacing information, and no independence is claimed for them.

Previsibility is not a substitute for splitting. It prevents leakage, the map at time t not containing y_t , and that is all it prevents. It does not make the transformed stream exchangeable: previsible strictly increasing maps that differ across time destroy exchangeability trivially, as when every second observation is shifted by a constant. Splitting does more. It freezes one map for calibration and test, so their scores stay exchangeable whenever the data are. The accurate statement is that emit-before-absorb guarantees prequential out-of-sample evaluation, and if the stream arriving at the fence-post rung is itself exchangeable, its sequential ranks are independent. The chains of Section 4 do not satisfy that hypothesis, and their certificate reads through the idealization already declared. Here the proposition separates the schedules. Split conformal’s p-values share one calibration set and are dependent. The prequential outputs are independent, and the price is early resolution: the map at time t has granularity $1/t$, hence the warmup of Section 2.

The certificate survives any further fixed monotone map, the probit included, because monotone maps preserve ranks. It is not automatically inherited through a downstream fit, by the second fact, and a chain that wants a terminal certificate ends with one more fence-post step on the final residuals. Full conformal is a third schedule, symmetrizing the whole chain over the data at the cost of refitting per candidate value (Vovk et al., 2005). We do not use it. An operator’s annotation is therefore a position and a clock.

For the economic series of Section 4 the exchangeable hypothesis of Proposition 3 is false, as it is throughout time-series conformal prediction, so the uniformity claims are conditional on model adequacy. Previsibility is not. It holds by construction, and every score in this paper is out-of-sample even where the guarantee hypothesis fails. The contracts so far assemble into the beginning of the type table the conclusion asks for:

operator	previsible	exch.-preserving	invertible	certificate
fence-post, discrete	yes	indep. seq. ranks	no	exact at grid levels
gaussianize, full grid	yes	not claimed	yes	exact on all n rank levels, $1/(n+1)$ globally
gaussianize, implemented	yes	not claimed	yes	exact on retained levels (distinct input, guard inactive), Δ_n plus guard displacement
fixed monotone link	yes	yes	yes	transparent
EMA location, EWMA scale	yes	no	given state	none
frozen split-fitted model	yes, coarse clock	cal/test pairs	varies	none
refit on the stream	yes	no	given state	none, blocks pass-through

The middle column is the working hypothesis of every chain in Section 4: location and scale do not preserve exchangeability, so the fence-post rung downstream of them operates under an idealization, which the contracts state rather than hide.

The fourth fact prices the choice of coordinates. The companion paper’s transform-pooling theorem states that pooling residuals R after an invertible state-dependent change of coordinates ϕ_X has irreducible log-score gap $I(\phi_X(R); X)$, the mutual information between the transformed residual and the state X on which its true conditional law depends: the Jacobians of the transform cancel, and only the coordinates’ ability to remove conditional heterogeneity matters (Cotton, 2026b). The gaussianize operator is a data-dependent monotone map, measurable with respect to its own past state, and piecewise C^1 , so conditional on the state the theorem’s hypotheses hold. The grammar view adds only this: since operators compose, the coordinates in which one pools are themselves a product of the chain, and the gap is a property of the whole chain up to the pooling point.

The operator’s ancestry is long. The branches are not one idea. They transform different objects for different reasons, and what they share is the trunk: push an observation through a distribution function and work on the flat scale.

The empirical distribution function is Galton’s ogive (Galton, 1875). Plotting positions place ordered observations at chosen probabilities, from Hazen’s $(i - \frac{1}{2})/n$ onward (Hazen, 1914), with $i/(n + 1)$ being Weibull’s position (Weibull, 1939), and the probability integral transformation received its name and first systematic treatment from Pearson (1938). Normal scores, the replacement of ranks by standard-normal quantiles, are the gaussianize map’s direct ancestor (van der Waerden, 1952). That estimation spoils the transform’s exactness has been known since David and Johnson (1948), and the positional fact above is its descendant.

Rosenblatt’s chaining through conditional distributions whitens any process (Rosenblatt, 1952), and normalizing flows are its learned, offline descendants (Tabak and Vanden-Eijnden, 2010; Papamakarios et al., 2021). Parametric monotone Gaussianizers run from Box and Cox (1964) to Yeo and Johnson (2000). Quantile mapping in the geosciences is the same move under another name (Panofsky and Brier, 1968). Transformed residuals as model criticism begin with Cox and Snell (1968), and the PIT as a forecast-evaluation instrument with Diebold et al. (1998). The prequential stance that all of this paper’s scoring obeys is Dawid (1984).¹

Modeling after marginal Gaussianization is classical outside conformal prediction too. Chen and Fan (2006) estimate an unknown time-series marginal by the rescaled empirical distribution with denominator $n + 1$, model temporal dependence with a parametric copula on the probability

¹A fuller annotated timeline is maintained at <https://skaters.microprediction.org/heritage.html>.

scale, and invert the fitted transition law for conditional forecasts. Meta-Gaussian hydrologic processors place the normal quantile transform inside a Gaussian predictive model and map back (Bogner et al., 2012). Autoregressive transformation models write the conditional law as a monotone transformation of a fixed base distribution (Rügamer et al., 2023). The distinction here is not the transform, model, inverse architecture. It is the finite-sample grid certificate attached to the empirical operator, the regularity debit charged against it, the online clock and state semantics, and score-based pooling over composed chains.

The transform also ran as a market before it ran here as an operator. The microprediction platform’s z-streams solicited competitive real-time predictions of PIT-transformed residual streams (Cotton, 2022), chaining solicitations so that one crowd predicted the residuals of another’s forecasts. That is interior composition as a mechanism, and the algebra of such mechanisms is developed in Cotton (2026a) and Cotton (2026c). The grammar of this paper is the single-forecaster form of the same design.

Against this ancestry the fence-post operator’s distinction is narrow and real: at its position, on exchangeable input, its ranks carry a finite-sample certificate.

4 Measurements

4.1 Design

The corpus, scoring conventions, and evaluation protocol follow the companion paper (Cotton, 2026b). We use 701 FRED change series with at least 600 observations and repeat fraction below five percent. The series are latest-vintage changes, so the claims concern sequential modeling of economic-shaped sequences, not operational real-time forecasting, where revisions and publication lags would enter. Every forecaster runs prequentially: at each time it issues a one-step predictive density from data strictly before that time, then sees the observation and updates. Scoring starts after a burn-in of 300 observations. The predictive in the original coordinates is the finite Gaussian mixture produced by the nine-particle inverse. That implemented mixture, blended with weight 0.01 toward an expanding-window Gaussian reference, is scored exactly, with no clipping or flooring in the reported nats. CRPS is computed in closed form from the Gaussian-mixture predictive. Per-series CRPS is normalized by the location baseline defined next, so scales are comparable across series.

The bare leaf, six single chains, and three pools are compared. All chains end in a Gaussian leaf with online variance, and all transforms update online.

config	chain
A	location (EMA) $\rightarrow$ leaf
B	location $\rightarrow$ scale (EWMA standardize) $\rightarrow$ leaf
C	location $\rightarrow$ gaussianize $\rightarrow$ leaf (rank-transform chain)
D	location $\rightarrow$ scale $\rightarrow$ gaussianize $\rightarrow$ leaf (normalized rank-transform chain)
E	location $\rightarrow$ scale $\rightarrow$ gaussianize $\rightarrow$ AR(1) $\rightarrow$ leaf (refit through)
F	location $\rightarrow$ scale $\rightarrow$ AR(1) $\rightarrow$ leaf (same refit, no gaussianize)
G	likelihood-weighted pool over {leaf, A, B, C, D, E, F}
H	likelihood-weighted pool over {leaf, A, B, F} (G minus the gaussianize chains)
I	the pool of G with kernel-smoothed gaussianize

Configs C and D are the conformal production rule written as a chain: transform, apply the fence-post empirical map, stop, and read the predictive off the inverse. Config E continues

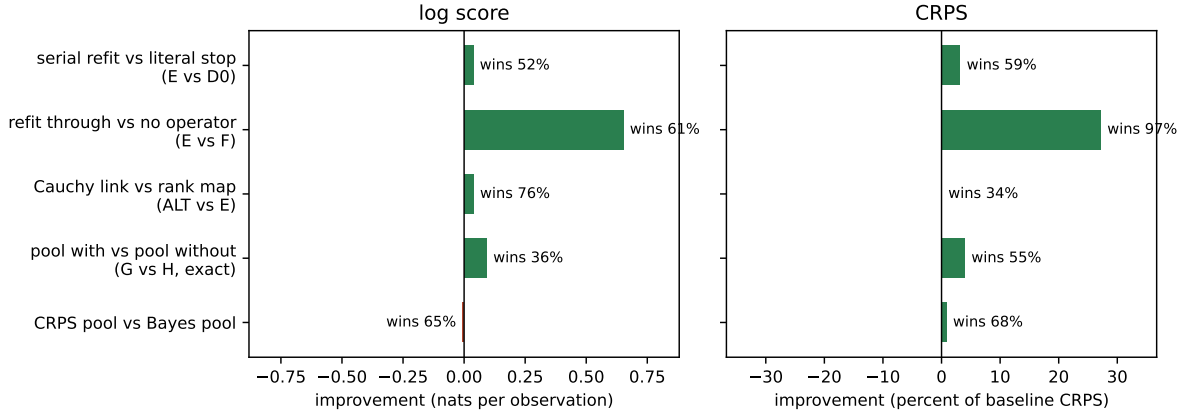


Figure 1: Five comparisons on the 572-series campaign corpus, under both scoring rules, with the repaired affine scale rung. Bars to the right favor the first-named config. Log score improvement is in nats per observation. CRPS improvement is the mean per-series ratio, expressed as the percent cut. Win rates are per-series.

through the operator. The pools use the same likelihood-weighted online ensembling as the companion paper’s forecaster, with a complexity penalty by chain depth.

A second campaign adds the decomposition ladder and robustness grids reported below. Config D0 is config D with its fitted leaf replaced by the fixed $N(0, 1)$ law, the terminal conformal forecast up to the nine-particle inverse representation, and a full-grid variant retains every knot. Alternative fixed links replace gaussianize in the E position. Guard and resolution grids, an iid null suite, and a CRPS-weighted pool complete the set. The refetch recovers 572 of the 701 series, mostly dropping short monthly ones, and the campaign runs use those, with percentile bootstrap intervals resampling series. The dropped series are mostly the short monthly ones. The corpus can always grow. Figure 1 summarizes the comparisons behind the paper’s three findings.

4.2 Stopping and inserting are different questions

In conformal practice the empirical map is terminal. One fits a model, ranks the residuals, and quotes the ranked law as the predictive. The fence-post step is where inference ends, and the design assumes the residuals arrive exchangeable, with nothing left in them worth modeling. Config D is that pattern written as a chain up to one caveat: its terminal leaf is a fitted unconditional Gaussian. Config D0 removes the caveat by quoting the fixed $N(0, 1)$ law at the operator’s output. Its scored predictive is the nine-particle representation of that law’s pullback, so it is the terminal conformal forecast up to inverse representation. Config E reads the same operator differently: its output is a coordinate system, and whatever structure the model missed is easier to fit in those coordinates. The two readings are testable against each other. For the exact sequential ranks, exchangeable input leaves no serial rank signal for a refit to find, by Proposition 3. The interpolated coordinates carry spacing information besides, and an estimated AR(1) brings estimation noise of its own, so the comparison is informative rather than clean.

The ladder, on the 572-series campaign corpus, separates three prices. Knot compression is

free: the full-grid variant differs from 39-knot D0 by 0.003 nats, compressed ahead. Refitting the terminal leaf hurts: D scores 0.011 nats below D0, winning 19 percent of series, because the operator has already standardized its output and a fitted variance adds estimation noise. Serial refitting beats the literal stop on average: E improves on D0 by 0.038 nats (interval 0.027 to 0.052) and cuts the mean per-series CRPS ratio by 3.2 percent, winning 59 percent of series on CRPS. The margin is skewed. The median log-score difference is a thousandth of a nat with wins on 52 percent of series, and the gains concentrate where serial structure is strong. On the iid null suite of Section 4.4 the same comparison runs the other way, by 0.011 to 0.051 nats depending on the null, which is the estimation cost of refitting when there is nothing to find, and it makes the corpus margin read as signal.

Under the log score the oracle version of this margin is the predictive mutual information remaining in the transformed stream, the quantity the companion paper’s ledger prices (Cotton, 2026b). The measured margin is the part of it an AR(1) captures, net of estimation cost. Serial structure survives the transform, and exploiting it is a skewed trade rather than a uniform one: the conformal stopping point is defensible series by series and beaten on the corpus.

A synthetic experiment shows the mechanism in isolation. Let $z_t = 0.8z_{t-1} + \varepsilon_t$ and observe $y_t = \text{sign}(z_t)|z_t|^{1.5}$, a monotone warp of an AR(1), which obscures the linear representation without removing the dependence. Stopping at gaussianize scores -2.27 nats and 1.24 CRPS. Refitting AR(1) through it scores -2.00 and 0.95 , beating both the stop and the same AR chain without the operator ($-2.13, 1.04$). When the downstream learner is well specified only in the empirical map’s coordinates, passing through the operator beats both stopping and applying the same learner in the original coordinates.

Stopping is one question, and inserting is another. E versus D0 asks whether the rank transform is a good architectural stopping rule. On average the answer is no. On the median series it is a coin flip. The operator’s incremental value is E versus F, the same refit without the operator. E beats F by 0.65 nats on average (interval 0.54 to 0.76), with a median of 0.06, wins on 61 percent of series, and wins the per-series CRPS comparison on 97 percent. Without a bounding rung, heavy-tailed coordinates reach the serial model and its leaf raw, and the worst series destroy the chain. In this chain the operator is not decoration. It is the difference between a usable pipeline and a fragile one.

The comparison is sensitive to the peers’ types, and one ablation shows how sharply. Replace the affine scale rung by a self-normalized one, dividing each emission by a running standard deviation that already contains it. The emitted coordinate is then bounded by construction, the chain without the operator inherits the bound, and the measured value of inserting the operator collapses to a small negative number. Bounding the coordinate is most of what the rank map supplies, and any rung that bounds on the side absorbs the operator’s credit. A mis-typed rung, one whose implemented map is not the map its inverse assumes, can misprice an operator two positions away. The type contracts are what the measurement of any one operator silently assumes about all the others.

Part of every particle-scored number is representation rather than operator. The exact-pullback runner scores the true change of variables, $\log q(m(y)) + \log m'(y)$, with analytic Jacobians and no particles. The nine-particle representation costs about 0.09 nats on the gaussianize chains, roughly constant across configs, and exactly zero on F, whose chain is affine and whose mixture pullback is already exact. Scored exactly on both sides, E beats F by 0.74 nats and beats D0 by 0.032, so the ladder’s conclusions are about the operator, not about its representation.

The alternative links complete the picture. A fixed Cauchy link in the same position beats

F by 0.69 nats, and it beats the rank map itself by 0.039 nats on average, winning 76 percent of series, while the rank map wins the per-series CRPS comparison against it on 66 percent. A sinh-arcsinh link at $\delta = 0.5$ wins 83 percent of series against the rank map and loses the mean to its own tail blowups, and the Yeo–Johnson links, which do not bound the tails at all, are destroyed with F. What the corpus rewards is heavy-tailed monotone bounding. The rank map supplies it adaptively, with a certificate. A fixed Cauchy link supplies it a shade better on average, with no certificate at all.

4.3 Nesting limits the cost of admission

The raw conformal pattern C is a reasonable forecaster on this corpus, better than the location baseline on average under both scores. It is also, on a minority of series, a bad choice. Config G, the pool over all seven chains, gaussianize chains included, beats the fixed pattern C on 87 percent of series by log score, by 0.163 nats on average. On the twenty series where C is worst, with relative CRPS 1.018, the pool sits at 0.824 on the same series. The pool walks away from the production rule where it underperforms.

Admitting the pattern now buys a real gain, and the shape is still lopsided. The exact Bayes pool G beats its sub-pool H, the same pool with the gaussianize chains removed, by 0.093 nats per observation on average (interval 0.072 to 0.115) and by 4 percent of the mean per-series CRPS ratio. The median difference is zero to four decimals and G wins 36 percent of series by log score: on most series the admitted chains change nothing, and where the repaired F-chains are fragile they change everything. Frequent negligible differences buy occasional rescues. Under the log score this shape is not luck. A standard Bayes-mixture regret inequality applies, and its interest here is what the subset represents: the conformal chains as a candidate block whose admission has a price.

Proposition 4 (Nesting is cheap under the log score). *Let the pool predictive be the Bayes mixture over chains with prior π , and let q_t be the predictive of the renormalized sub-mixture over any subset Q of chains, with prior mass π_Q . Then for every realization $y_1, \dots, y_T$,*

$$-\sum_{t=1}^T \log p_t(y_t) \leq -\sum_{t=1}^T \log q_t(y_t) + \log \frac{1}{\pi_Q}.$$

Proof. For a Bayes mixture the cumulative predictive likelihood telescopes to the prior-weighted sum of the chains’ likelihoods, $\prod_t p_t(y_t) = \sum_c \pi_c \prod_t p_{t,c}(y_t)$. Dropping the chains outside Q and renormalizing, $\prod_t p_t(y_t) \geq \pi_Q \sum_{c \in Q} (\pi_c / \pi_Q) \prod_t p_{t,c}(y_t) = \pi_Q \prod_t q_t(y_t)$. Take logarithms. $\square$

Take Q to be the sub-pool without the gaussianize chains. The total price of admitting them is then at most $\log(1/\pi_Q)$ nats, however the data falls, and the per-observation price vanishes. The mass π_Q is the prior mass of the retained block, not of the admitted one. The reported pools are exact Bayes mixtures, so the proposition applies to them verbatim: no likelihood clipping, no cross-expert mixture pruning, the reference blend inside the weight updates, and the depth prior applied once. The operator’s $|z| \leq 8$ learner-input channel stays active inside each expert, as everywhere in the paper. The production ensemble instead tempers its likelihood update at rate 0.8 and applies its depth penalty per step, for adaptivity under regime change. Its comparison does not inherit the exact bound, and it shows the same picture, the tempered G ahead of its H by 0.078 nats. The runner resets the mixture weights to the depth prior at the start of the scored window, because the bound’s prior-mass penalty controls the loss from

initialization, not from an arbitrary suffix: scoring after a burn-in that already reweighted the pool would need the realized posterior mass in place of the prior. With the reset, the sub-pool prior mass $\pi_Q = 0.573$ makes the worst case 0.56 nats in total for the reported scores, however long the series run, and the realized comparison, 0.093 nats per observation in G’s favor, sits far on the profitable side of it.

No such bound is available to us under CRPS, and there the claim stays empirical. The weights update by likelihood, so the log score does the choosing and CRPS only evaluates the chosen blend. The matching experiment is a CRPS-weighted exponential pool over the same seven chains with the same prior. Its comparison with the exact Bayes pool is mixed and close. The CRPS-weighted pool wins 65 percent of series by log score and 68 percent by CRPS, with median CRPS ratio 0.99, while its mean log score is 0.006 nats lower with an interval spanning zero. The admission rule is itself a choice within the grammar, and neither weighting dominates the other here.

This is the practical content of the grammar view. A fixed conformal method is a commitment, and the companion paper prices what the commitment can cost. A grammar that contains the method as one rule converts the commitment into a weighted vote, and the vote is revised online by the scoring rule. Nesting does not avert disasters on this corpus, because the guarded operator does not produce disasters. An exact Bayes pool trims the bad tail at a bounded cumulative price, vanishing per observation.

4.4 Robustness

Three checks bound the sensitivity of these findings. An iid null suite runs the ladder when there is nothing to find. The kinds are Gaussian, Student t_3 , lognormal skew, and a two-component mixture, forty replicates each. Against the literal stop the refit loses on every null, by 0.011 to 0.051 nats depending on kind, which is the estimation cost of fitting absent structure and is the sign the corpus reverses. Against the fitted leaf the refit gains up to 0.023 nats on the skewed and mixture nulls, so part of any E-versus-D margin is adaptation rather than serial structure, and the stopping comparison is E versus D0 for exactly this reason. A guard grid crossing slope caps $\{10, 20, 40\}$ with clamps $z_{\max} \in \{6, 8, 10\}$ on a hundred-series subsample moves config E by 0.003 nats end to end, so the guard constants are not doing the work. A resolution grid on the same subsample finds the knot count flat in the point estimates: 79, 159, and the full grid all sit within 0.003 nats of 39, none ahead of it. On a hundred series those intervals cannot exclude effects of the size the particle count shows. Raising the inverse from nine particles to nineteen is worth 0.016 nats, and going further is not, which is the subsample’s view of the 0.09-nat representation price the exact-pullback runner measures in full. The kernel-smoothed C^1 variant of the operator tracks the linear one, 0.003 nats below it on the refit chain, so smoothness is not where the value lives. The certificate’s resolution parameter is not the empirical bottleneck. The inverse’s resolution is, mildly.

5 One family

The grammar view collapses a scattered vocabulary. Platt scaling (Platt, 1999), isotonic calibration (Zadrozny and Elkan, 2002), temperature scaling (Guo et al., 2017), quantile recalibration (Kuleshov et al., 2018), conformal predictive systems (Vovk et al., 2019), and distributional conformal prediction (Chernozhukov et al., 2021) can all, in univariate probabilistic forecasting, be represented as monotone recalibration operators on a scalar predictive, score, or probability

coordinate, differing in what they fit, what they guarantee, and where in the chain they sit. The forecasting literature’s calibration-sharpness tradition (Gneiting et al., 2007; Gneiting and Raftery, 2007) supplies the objective that orders them. In the grammar they are not competing philosophies. They are operators with types, and a pool over chains can hold several at once.

Operator	Coordinate	Map	Invertible	Certificate at output
Platt	score	fitted sigmoid	yes	none
Isotonic	score	fitted monotone step	no	none
Temperature	logit	fitted scale	yes	none
Quantile recal.	PIT	fitted empirical CDF	a.e.	asymptotic
CPS	score	rank map	at knots	finite-sample, at position
DCP	PIT	rank map	at knots	finite-sample, at position
Fence-post (here)	residual	interpolated rank map	yes	finite-sample, at position

Two wrapper lines make the composition explicit. Conformal calibrators (Vovk et al., 2020) apply the rank map to the output of an arbitrary predictive system, and modular conformal calibration (Marx et al., 2022) unifies the recalibration family the table lists. Both are one level of composition, and in both the map is terminal by construction.

The time-series conformal systems nest as production rules too. Sequential predictive conformal inference (Xu and Xie, 2023) replaces the unconditional rank rung with a fitted conditional quantile model on the residual stream: the same slot, a different operator, the certificate traded for sharpness. Ensemble batch prediction intervals (Xu and Xie, 2021) keep the empirical rung and give it a sliding-window clock. Adaptive conformal inference (Gibbs and Candès, 2021) composes an operator after the rank map, a feedback controller that moves the quoted level in response to coverage errors, and conformal PID control (Angelopoulos et al., 2023) adds a scorecaster, a serial forecaster fitted on the score stream that feeds the controller. Flow-based conformal regression (Colombo, 2024) fits an invertible reshaping upstream and applies the rank map to its output, our location and scale rungs with a richer transform.

Each system holds the rank map at a fixed position and reads out a set, an interval, or a distribution it then quotes. The one operator that literature composes after the map is a controller on the quoted level, driven by coverage. Among the conformal systems above, we have not found one that fits a predictive-density model on the rank map’s observation-coordinate output, carries predictive densities back through its inverse, or scores the map as one operator among peers under a proper scoring rule. Those are the moves this paper adds, and the nesting is the point: every system above is a chain the pools of Section 4 could hold.

A probabilistic lifting of the same geometry appears in Snell and Griffiths (2025). If $T_{(i)} = F(Y_{(i)})$ are the latent probability levels of the ordered observations, the spacings of the $T_{(i)}$ are $\text{Dirichlet}(1, \dots, 1)$, and the plotting positions $i/(n+1)$ used here are their means. The present operator fixes the mean positions and obtains a deterministic coordinate transform. Snell and Griffiths retain the full spacing distribution and push a loss integral through it, and standard conformal risk control is recovered by replacing the random spacings with their expectations. In grammar terms theirs is a measure-valued operator on the loss coordinate, emitting a distribution over risk bounds rather than a transformed observation, and it is complementary to the observation-coordinate transforms studied here. The two demotions agree: conformal prediction is what remains of an empirical rank geometry after one functional of it is retained and the rest discarded.²

²One typing caution. The Dirichlet law is a sampling statement about the latent spacings, and reading it as

Conformal prediction keeps a distinguished place in this family. The fence-post operator is the member whose output carries a finite-sample, distribution-free guarantee at its position in the chain (Proposition 2). No fitted recalibration offers that. The grammar view does not diminish the guarantee. It names what the guarantee attaches to, and it frees the forecaster to keep transforming afterward.

6 Conclusion

Conformal prediction is one production rule in a grammar of composable transforms: model, residual, fence-post empirical map, inverse. Interpolated, the fence-post map is an ordinary invertible online transform, and the rule’s distinguishing stop, quoting the empirical law where it stands, becomes a choice the scoring rule can overrule. On economic series, a serial model added after the operator outscores the literal terminal stop on average. Inserted into a chain whose scale rung is honestly affine, the operator is load-bearing under both scoring rules, and its measured value collapses only when some other rung bounds the coordinate on the side. A pool that admits the pattern pays a bounded price and collects a real gain. A fixed Cauchy link supplies the same bounding a shade better on average, and carries no certificate.

The operator is exceptional as a certificate and ordinary as an architecture, and the certificate it carries is priced in its computational budget: retained-knot resolution, tie mass, and guard displacement set the calibration error. The typing question is the theory this view opens. An operator’s guarantee attaches to a position in a chain. Which properties survive composition is a discipline we have only begun here, with one operator, a few propositions, and the first rows of a table, and the rest of the table is open.

A The certificate, priced

The slope guard’s declaration has a price, and three objects need separating. The exact-knot interpolant is what the propositions below certify. The slope-guarded map is what the experiments run. Its cap is enforced by lowering any knot whose segment is too steep. When it binds, the map no longer passes through the displaced order statistics, and retained-level exactness is suspended there. The clamped emission is a third object, a winsorized channel to downstream state and not part of the invertible map. The nine-particle mixture representation of the pullback is a fourth: the campaign scores that mixture exactly, which is exact scoring of an approximate distribution, not exact change of variables, and its representation error is separate from all three prices above. Remark 10 prices the middle one.

Proposition 5 (Grid calibration survives interpolation). *Let $Y_1, \dots, Y_{n+1}$ be exchangeable and almost surely distinct. Let m be any strictly increasing map, interpolation and tails arbitrary, that passes through the knots: $m(y_{(i)}) = z_{(i)}$ at the order statistics $y_{(1)} < \dots < y_{(n)}$ of $Y_1, \dots, Y_n$. Then*

$$\Pr\{m(Y_{n+1}) \leq z_{(i)}\} = \frac{i}{n+1}, \quad i = 1, \dots, n.$$

a data-conditional posterior requires an argument about the prior over F that the sampling statement does not supply. Sampling-law, posterior-law, and prequential certificates are different types, and the table should keep them apart.

Proof. Monotonicity gives $\{m(Y_{n+1}) \leq z_{(i)}\} = \{Y_{n+1} \leq y_{(i)}\}$. By exchangeability and distinctness the rank of Y_{n+1} among all $n+1$ values is uniform on $\{1, \dots, n+1\}$, and $Y_{n+1} \leq y_{(i)}$ holds for exactly the i lowest ranks. $\square$

There is no need to insist on exactness off the grid, because the relaxation is uniformly small and assumption-free. Call an operator δ -conformal if, on exchangeable almost surely distinct input, its output U satisfies $\sup_u |\Pr\{U \leq u\} - u| \leq \delta$.

Proposition 6 (Interpolation is $1/(n+1)$ -conformal). *In the setting of Proposition 5, write $U = \Phi(m(Y_{n+1}))$ for the probability-scale output of any interpolated map through the knots. Then*

$$\sup_{u \in (0,1)} |\Pr\{U \leq u\} - u| \leq \frac{1}{n+1},$$

whatever the interpolation, the tails, and the continuous input law. Equivalently, $m(Y_{n+1})$ is within Kolmogorov distance $1/(n+1)$ of standard Gaussian.

Proof. For u between consecutive grid levels $i/(n+1)$ and $(i+1)/(n+1)$, monotonicity of the distribution function and Proposition 5 trap $\Pr\{U \leq u\}$ in $[i/(n+1), (i+1)/(n+1)]$, the same cell that contains u . Two points of one cell differ by at most $1/(n+1)$. The edge cells are the same argument with endpoints 0 and 1. $\square$

The bound generalizes to the implemented operator, which does not keep every knot.

Proposition 7 (The certificate scales with the retained resolution). *Let $0 = r_0 < r_1 < \dots < r_K < r_{K+1} = n+1$, and let m be any strictly increasing map with $m(y_{(r_j)}) = \Phi^{-1}(r_j/(n+1))$ at the retained order statistics, interpolation and tails arbitrary. Then $U = \Phi(m(Y_{n+1}))$ satisfies*

$$\sup_{u \in (0,1)} |\Pr\{U \leq u\} - u| \leq \Delta_n := \max_{0 \leq j \leq K} \frac{r_{j+1} - r_j}{n+1}.$$

If each retained knot's value sits within e stream ranks of its nominal rank, the bound degrades to $\Delta_n + e/(n+1)$. The hypothesis of Proposition 5, exchangeable and almost surely distinct input, is in force.

Proof. The argument of Proposition 5 gives exactness at each retained level $r_j/(n+1)$. Between consecutive retained levels, monotonicity traps $\Pr\{U \leq u\}$ and u in one retained cell, of width at most Δ_n . For the rank-error term, the hypothesis is a bracketing contract: each interpolation knot $\bar{y}_j$ satisfies $y_{(r_j-e)} \leq \bar{y}_j \leq y_{(r_j+e)}$ almost surely. Event inclusion then traps $\Pr\{Y_{n+1} \leq \bar{y}_j\}$ between $(r_j - e)/(n+1)$ and $(r_j + e)/(n+1)$, so the level error at a knot is at most $e/(n+1)$, which adds to the cell width. $\square$

The full map is the case $\Delta_n = 1/(n+1)$. For the implemented operator the accounting has two regimes. While the slope guard is inactive the retained knots are exact order statistics, $e = 0$, and after warmup the certificate is roughly one part in forty everywhere, exact at the retained levels. When the guard binds it lowers the offending knots, the map no longer passes through the displaced order statistics, and the displacement enters e through the bracketing contract. The displacement need not be small, as Remark 10 shows, so the guarded operator is not Δ_n -conformal unconditionally. The e term is also the price a future bounded-memory compaction would pay.

Proposition 7 needs a budget e declared for every possible sample. A guard that inspects the realized sample and reports the smallest feasible displacement has produced a data-dependent quantity, selected by the same data that built the map, and it cannot be substituted into the unconditional bound. Two repairs work. Declare fixed budgets e_0 and Δ_0 in advance, with a declared fallback when the slope budget is not achievable within them, and the deterministic certificate $\Delta_0 + e_0/(n+1)$ stands. Or condition on the multiset and take worst cases over the held-out slot. The second repair is the next statement, and it covers ties as well.

Proposition 8 (Leave-one-out certificate). *Let $Y_1, \dots, Y_N$ be exchangeable, $N = n+1$, and let the knot construction depend on its calibration values only through their multiset, as the sorted-buffer operator does after warmup. Condition on the multiset M of the N values. For each slot s , build the map m_s from the other n values, and suppose its retained knots satisfy the bracketing contract with rank displacement at most e_s and retained-cell width at most Δ_s . Let Δ^* and e^* be the maxima over slots, and $\beta^* = (b^* - 1)/N$ with b^* the largest tie-block size of M . Let S be the held-out slot, uniform given M by exchangeability, and $U = \Phi(m_S(Y_S))$. Then, almost surely,*

$$\sup_u |\Pr\{U \leq u \mid M\} - u| \leq B(M) := \Delta^* + \frac{e^* + 1}{N} + \beta^*,$$

and unconditionally $\sup_u |\Pr\{U \leq u\} - u| \leq \mathbb{E}[B(M)] \leq \text{ess sup } B(M)$.

Proof. Fix $u \in (0, 1)$. For slot s , let $a_s \leq Nu \leq b_s$ be the retained ranks of slot s 's grid adjacent to Nu , with the conventions $r_{s,0} = 0$ and $r_{s,K_s+1} = N$, so that $(b_s - a_s)/N \leq \Delta^*$, and this holds even when the grids differ across slots, as under the deletion guard. Set $a = \min_s a_s$ and $b = \max_s b_s$; then $a/N \geq u - \Delta^*$ and $b/N \leq u + \Delta^*$. For slot s the event $\{U \leq u\}$ is $\{Y_s \leq m_s^{-1}(\Phi^{-1}(u))\}$, and monotonicity places $m_s^{-1}(\Phi^{-1}(u))$ between slot s 's knots at ranks a_s and b_s , with the convention that the interval is unbounded on a side where the boundary rank is 0 or N . Removing one value moves order statistics by at most one position, $Y_{(k)} \leq y_{(k)}^{(s)} \leq Y_{(k+1)}$, so the bracketing contract traps those knot values between $Y_{(a-e^*)}$ and $Y_{(b+e^*+1)}$. At least $a - e^*$ of the N values are at most $Y_{(a-e^*)}$, and at most $b + e^* + 1 + (b^* - 1)$ are at most $Y_{(b+e^*+1)}$. The count of slots satisfying the event therefore lies between $a - e^*$ and $b + e^* + 1 + (b^* - 1)$, and dividing by N , $\Pr\{U \leq u \mid M\}$ differs from u by at most $B(M)$. Taking expectations gives the marginal statement. $\square$

Corollary 9 (Tie handling). *Drop the distinctness assumption. For the discrete rank, randomization restores exactness: ordering tied observations uniformly at random makes the rank of Y_{n+1} uniform on $\{1, \dots, n+1\}$ whatever the tie structure, so Proposition 2 survives ties unchanged. Interpolation admits no such fix, because tied values are numerically equal and re-ordering them does not change the map. For the interpolated operator with one knot per distinct retained value, Proposition 8 applies as stated: its tie term β^* is the price, and exactness at retained levels is lost. The corollary is the tied case of the proposition, and needs no separate proof.*

Remark 10 (The guard is a third price). Take an exchangeable vector of $N = 79$ distinct values, seventy-one at $0, 1, \dots, 70$ and eight near 10^9 , and hold one out. With 39 retained knots the top knot of the calibration map sits at the seventy-eighth calibration order statistic and $\Delta_n = 2/79$. The moving cap, at twenty times the central inverse slope, drags that knot from the high cluster down to roughly 671, between the clusters. At the retained level $78/79$ the probability that the held-out value falls below the displaced knot is $71/79$, an error of $7/79$, exceeding Δ_n . Retained

resolution and tie mass are therefore not the whole price of the certificate. Guard displacement is a third term, zero exactly while the guard sleeps.

Two guards compatible with the certificate are implemented as options. Deleting offending knots keeps the survivors at exact order statistics and enlarges the realized cell width. The smallest-displacement guard solves, for fixed e , the feasibility system

$$y_{(r_j-e)} \leq \bar{w}_j \leq y_{(r_j+e)}, \quad 0 \leq \bar{w}_{j+1} - \bar{w}_j \leq L(x_{j+1} - x_j),$$

with flat knots dropped for invertibility, and obtains the certified upper bound $\Delta_n + e/(n+1)$ with the smallest e feasible for the fixed retained grid. When flat knots are dropped the grid changes, so Δ_n in the bound is the width recomputed after the drop, and minimizing e alone need not minimize the total: the guard fitted exactly to the certificate would minimize $\Delta_n + e/(n+1)$ jointly. On this example the per-slot smallest displacement is $e = 7$ and the leave-one-out maximum of Proposition 8 is $e^* = 8$.

Three certificates must be kept apart. Budgets e_0 and Δ_0 declared in advance give a prospective certificate, available before the test value arrives. The leave-one-out maxima e^* and Δ^* give the conditional certificate of Proposition 8, and computing them needs the completed multiset and N reconstructions, so it is an ex-post transductive diagnostic. The single realized e_S and Δ_S that a guard reports on the forecasting path bound neither and are a diagnostic only. On the campaign corpus, replacing the moving cap by either guard moves mean log score by less than 0.0003 nats, with identical output on eight in ten series.

The corpus of Section 4 filters to repeat fraction below five percent rather than zero, so β_n is small and not vanishing there. The certificate is a function of knot resolution, sketch error, tie mass, and guard displacement: the guarantee is priced in the operator’s computational budget.

The discrete fence-post map and gaussianize-full both have global error at most $1/(n+1)$: the discrete map’s output is supported on the grid, so as a distribution on $[0, 1]$ it too sits within Kolmogorov distance $1/(n+1)$ of uniform, and only the full map is invertible. With distinct input and the guard inactive, the implemented retained-knot map is exact at retained levels with global error at most Δ_n . Under a guarded displacement with uniform budget e the bound is $\Delta_n + e/(n+1)$, and with data-dependent guards it is read through Proposition 8.

Regularity of empirical transforms has a separate lineage, and the guard is not new as an optimization. Write $x_j = \Phi^{-1}(r_j/(n+1))$ and $w_j = y_{(r_j)}$, so that the inverse empirical Gaussianizer is the monotone quantile transport $x_j \mapsto w_j$. Replacing w_j by nearby values $\bar{w}_j$ satisfying $0 \leq \bar{w}_{j+1} - \bar{w}_j \leq L(x_{j+1} - x_j)$ is Lipschitz isotonic regression, studied by Yeganova and Wilbur (2009), Agarwal et al. (2010), and Kakade et al. (2011), with the bounded-increment ancestry running back to isotonic optimization (Jewell, 1975). Paty et al. (2020) identify the two-sided version, lower and upper slope bounds together, as the one-dimensional case of regularized empirical optimal transport, recovered by piecewise-affine interpolation. The unguarded operator has close kin as well: the continuous invertible empirical distribution transform of Harar et al. (2022) is essentially the full-grid map without the conformal reading, and truncated normal scores go back to the nonparanormal (Liu et al., 2009).

So the guard is not offered as a new slope-constrained regression method, and the operator is not offered as a new transform. In transport language the inverse Gaussianizer is a one-dimensional empirical transport map and the guard is Lipschitz isotonic regularization of it. The role of the guard here is to declare an inverse sensitivity budget and to expose the finite-sample calibration price of enforcing it, through the rank-displacement term of Proposition 7. The canonical guard would be the least-squares projection onto the slope-constrained set, the

regularized transport of Paty et al. (2020). A guard fitted to this paper instead minimizes the rank displacement e subject to the budget, solving exactly the optimization the proposition prices. The operator implements both this and the deletion guard as options, and Remark 10 measures them.

B Reproducibility

Every number in Section 4 is produced by one of four runners in the `benchmarks` directory of the `skaters` repository. They are `gaussianize_chain.py` and `grammar_campaign.py`, with `grammar_exact_bayes.py` and `grammar_exact_pullback.py` described below. The original run is `gaussianize_chain.py`, with per-series results frozen in `gaussianize_chain.csv`, and it also produces the synthetic warp numbers. The campaign runner is described at the end of this appendix. The configuration is as follows. The location operator is an EMA transform with rate 0.1. The scale operator is an EWMA standardization with rate 0.05, emitting against the prior state so that the rung is the affine change of coordinates its inverse applies, with the first informative emission scaled to one as cold start. The AR(1) operator estimates its coefficient by recursive least squares with forgetting factor 0.99 and ridge 1.0. The gaussianize operator uses at most 39 knots at equal mass from the full sorted history of its inputs (exact order statistics before the slope guard, which can displace knots when it binds; the buffer is unbounded), a warmup of 30 observations with expanding-moment standardization before it, and a 9-particle equal-probability approximation for the inverse. The kernel-smoothed variant uses a Gaussian-kernel distribution function at Silverman bandwidth, represented by a monotone C^1 cubic with Fritsch–Butland tangents and inverted by Newton’s method. The guards are the inverse-slope cap at twenty times the interquartile inverse slope and the downstream-update clamp $|z| \leq 8$. The implementation returns the clamped value from the forward callable, so the software object is not itself injective: the invertible map and the bounded update channel are separated in the analysis, and a typed implementation would return them as two channels.

The pools are tempered likelihood mixtures over the candidate chains of the design table, with tempering rate 0.8, a complexity penalty of 0.005 per unit of chain depth applied to the online weights, and at most 20 mixture components. Scoring starts after a burn-in of 300 observations. The log score mixes each predictive with weight 0.01 toward an expanding-window Gaussian reference and is otherwise exact. CRPS is computed in closed form from the Gaussian-mixture predictive and normalized per series by config A. The corpus is 701 FRED change series with at least 600 observations and repeat fraction below five percent, listed in the frozen results file. Series were downloaded from FRED on 2 August 2026 with observation start 2005-01-01, and non-numeric observations are dropped at parse. The runner does not log guard activations separately. The one recorded activation episode is the warmup event described in Section 2.

The decomposition ladder, alternative links, null suite, guard and resolution grids, and CRPS pool are produced by `benchmarks/grammar_campaign.py` in the same repository, with per-series results frozen in the `grammar_campaign_*.csv` files. The campaign corpus is the 572 of the 701 series whose full history could be refetched on 3 August 2026. Config D0 ends in a fixed $N(0, 1)$ leaf. The fixed links use the same nine-particle inverse pushback as gaussianize: Yeo–Johnson at $\lambda \in \{0.5, 1.5\}$, $\sinh$ -arcsinh at $\delta \in \{0.5, 2\}$, and the Cauchy link $\Phi^{-1}(F_{\text{Cauchy}})$. The null suite draws series of length 1200 and forty replicates per kind. The CRPS pool uses exponential weights with rate 1 on cumulative CRPS and the exact-Bayes prior. The guard and resolution subsamples take every fifth series alphabetically, truncated to one hundred, a deterministic rule

that never samples the last seventy-odd series. The deletion-guard and smallest-displacement comparisons are the `delguard` and `emin` subcommands, using the operator’s `guard` option, on the full campaign corpus. The exact Bayes pools are produced by `grammar_exact_bayes.py`, which updates weights by exact blended likelihoods without clamping or pruning and reports CRPS from a pruned copy of the predictive, the log score being exact. Bootstrap intervals are percentile intervals resampling series, 2000 draws, seed 0.

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